Abinash Selvarasu

Software Engineer · Java, Spring Boot, Go · Chennai, India
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Summary

Software engineer building payment and regulatory integrations for banking clients with Java, Spring Boot, Kafka, and Oracle. First author of peer-reviewed Springer papers on deep-learning intrusion detection for industrial and cyber-physical systems. Outside work I build Go services and developer tools, including a local, privacy-first AI code reviewer.

Experience

Mindgate Solutions Jul 2025 – Present
Software Engineer *Chennai, India*

- Engineer on **VTransact**, a multi-country corporate tax-payment platform (see Projects).
- Developed an internal **cryptographic digital-signature generator** used to sign requests for a Middle East regulatory API integration.
- Built Java/Spring Boot **microservices** on WebSphere, using **Kafka** and **WebSphere MQ** for messaging and **Oracle DB**/PostgreSQL for persistence.
- Wrote **Spring Batch** data-migration tools and other internal utilities used day to day by the engineering team.

Projects

VTransact | *Java, Spring Boot, Kafka, WebSphere MQ, Oracle* 2025 – Present

- **Multi-country tax-payment portal** that lets a multinational bank's corporate customers pay taxes and government dues across countries from one platform.
- Integrates with **government and regulatory systems** in each country, including **multiple Nigerian payment channels** for tax collection and a Middle East regulatory API secured with **digitally signed requests**.
- Payment and status messages flow between services over **Kafka** and **WebSphere MQ**, with Oracle DB as the system of record.

Codist | *Go, Svelte, tree-sitter, SQLite, Ollama* 2026, v0.2.0

- Local AI code reviewer for Java/Spring Boot that reviews every commit with exact code checks, a local LLM, and behaviour runs; ships as a **single cross-platform Go binary**, split into 8 Go modules, and nothing leaves the machine.
- Parses Java with a pure-Go **tree-sitter** parser into class and call graphs, including **Feign calls between services**; rules such as a repository call inside a loop or seconds passed as milliseconds run on changed lines before the model.
- Grounds the LLM: every claim must **quote the code** it means or is dropped; issues are tracked across commits until fixed; a **logistic-regression risk model** trained on the repo's history picks which commits the model reads first.
- **Behaviour runs** call changed methods before and after a commit in separate class loaders and diff the results; on a 24-bug Spring Boot benchmark, Codist flagged the right line for **23 of 24** bugs.

Publications

- **S. Abinash**, N. Srivatsan, S. K. Hemachandran, S. Priyanga. *HGCNN-LSTM: A Data-driven Approach for Cyberattack Detection in Cyber-Physical Systems.* SN Computer Science (Springer), vol. 6, 2025.
- **S. Abinash**, V. S. Shankar Sriram, K. Krithivasan, S. Priyanga. *HATT-MLPNN: A Hybrid Approach for Cyber-Attack Detection in Industrial Control Systems Using MLPNN and Attention Mechanisms.* ATIS 2024, CCIS (Springer), 2024.
- **S. Abinash**, S. Priyanga, H. Aishwaryaa. *HyGAN-AD: A Hypergraph-based GAN Framework for Cyberattack Detection in Critical Infrastructures.* Lecture Notes in Networks and Systems (LNNS, Springer).
- **S. Abinash**, S. Priyanga. *HyperCoar-DAM: A Robust Intrusion Detection System for Enhancing IIoT Networks Protection.* International Journal of Advances in Engineering Sciences and Applied Mathematics.

Technical Skills

Languages: Java, Python, Go, TypeScript, Kotlin, SQL, Lua
Backend: Spring Boot, Spring MVC, Spring Batch, Lombok, Gin, Echo, REST APIs, Microservices
Data & Messaging: Oracle DB, PostgreSQL, libSQL, Apache Kafka, IBM WebSphere MQ
Frontend & Tools: React, Next.js, Svelte, Qwik, Tailwind CSS, Git, Docker, Linux, Cloudflare Workers

Education

SASTRA Deemed University Thanjavur, India
B.Tech, Computer Science and Engineering *2021 – 2025*